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1. Incorrect JOIN Syntax Returning Duplicate Results

```
mysql> SELECT E.Name,D.DeptName  
-> FROM Employee E  
-> JOIN Department D;
```

Name	DeptName
Raj	Finance
Raj	HR
Raj	IT
Arun	Finance
Arun	HR
Arun	IT
Divya	Finance
Divya	HR
Divya	IT
Meena	Finance
Meena	HR
Meena	IT
Karan	Finance
Karan	HR
Karan	IT

15 rows in set (0.02 sec)

```
mysql> SELECT E.Name,D.DeptName  
-> FROM Employee E  
-> INNER JOIN Department D  
-> ON E.DeptID=D.DeptID;
```

Name	DeptName
Raj	IT
Arun	IT
Divya	HR
Meena	Finance
Karan	Finance

5 rows in set (0.00 sec)

2. Missing Correlation in Subquery

```
mysql> SELECT Name
-> FROM Employee E
-> WHERE Salary >
-> (
-> SELECT AVG(Salary)
-> FROM Employee
-> );
+-----+
| Name |
+-----+
| Arun |
| Meena |
+-----+
2 rows in set (0.00 sec)
```



```
mysql> SELECT Name
-> FROM Employee E
-> WHERE Salary >
-> (
-> SELECT AVG(Salary)
-> FROM Employee
-> WHERE DeptID=E.DeptID
-> );
+-----+
| Name |
+-----+
| Arun |
| Meena |
+-----+
2 rows in set (0.00 sec)
```

3. Incorrect SQL View

```
mysql> CREATE VIEW EmployeeView AS
-> SELECT *
-> FROM Employee
-> GROUP BY DeptID;
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE VIEW EmployeeView AS
-> SELECT EmpID,
-> Name,
-> Salary,
-> DeptID
-> FROM Employee;
ERROR 1050 (42S01): Table 'EmployeeView' already exists
```

4. Incorrect Relational Algebra Natural Join

```
mysql> SELECT *
-> FROM Employee
-> NATURAL JOIN Department;
+-----+-----+-----+-----+-----+-----+-----+
| DeptID | EmpID | Name  | Salary | DeptName | Location | ManagerID |
+-----+-----+-----+-----+-----+-----+-----+
| 10     | 101   | Raj   | 50000.00 | IT       | Chennai  | 1         |
| 10     | 102   | Arun  | 65000.00 | IT       | Chennai  | 1         |
| 20     | 103   | Divya | 45000.00 | HR       | Bangalore | 2         |
| 30     | 104   | Meena | 70000.00 | Finance  | Chennai  | 3         |
| 30     | 105   | Karan | 55000.00 | Finance  | Chennai  | 3         |
+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT *
-> FROM Employee
-> JOIN Department
-> ON Employee.DeptID=Department.DeptID;
+-----+-----+-----+-----+-----+-----+-----+
| EmpID | Name  | Salary | DeptID | DeptID | DeptName | Location | ManagerID |
+-----+-----+-----+-----+-----+-----+-----+
| 101   | Raj   | 50000.00 | 10     | 10     | IT       | Chennai  | 1         |
| 102   | Arun  | 65000.00 | 10     | 10     | IT       | Chennai  | 1         |
| 103   | Divya | 45000.00 | 20     | 20     | HR       | Bangalore | 2         |
| 104   | Meena | 70000.00 | 30     | 30     | Finance  | Chennai  | 3         |
| 105   | Karan | 55000.00 | 30     | 30     | Finance  | Chennai  | 3         |
+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

5. Referential Integrity Violation

```
mysql> INSERT INTO Employee
-> VALUES(106,'John',60000,50);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('simats`.`employee`, CONSTRAINT `employee_ibfk_1` FOREIGN KEY (`DeptID`
`) REFERENCES `department` (`DeptID`))
mysql> INSERT INTO Department
-> VALUES(50,'Marketing','Madurai',2);
Query OK, 1 row affected (0.03 sec)

mysql>
mysql> INSERT INTO Employee
-> VALUES(106,'John',60000,50);
Query OK, 1 row affected (0.03 sec)
```

6. Nested Query Optimized Using JOIN

```
mysql> SELECT Name
-> FROM Employee
-> WHERE DeptID IN
-> (
-> SELECT DeptID
-> FROM Department
-> WHERE Location='Chennai'
-> );
+-----+
| Name |
+-----+
| Raj  |
| Arun |
| Meena |
| Karan |
+-----+
4 rows in set (0.02 sec)

mysql> SELECT E.Name
-> FROM Employee E
-> JOIN Department D
-> ON E.DeptID=D.DeptID
-> WHERE D.Location='Chennai';
+-----+
| Name |
+-----+
| Raj  |
| Arun |
| Meena |
| Karan |
+-----+
4 rows in set (0.00 sec)
```

7.Incorrect GROUP BY

```
mysql> SELECT DeptID,  
-> Name,  
-> AVG(Salary)  
-> FROM Employee  
-> GROUP BY DeptID;  
ERROR 1055 (42000): Expression #2 of SELECT list is not in GROUP BY clause and contains nonaggregated column 'simats.Employee.Name' which is not functionally dependent on columns in GROUP BY clause; this is incompatible with sql_mode=only_full_group_by  
mysql> SELECT DeptID,  
-> AVG(Salary)  
-> FROM Employee  
-> GROUP BY DeptID;  
+-----+-----+  
| DeptID | AVG(Salary) |  
+-----+-----+  
| 10 | 57500.000000 |  
| 20 | 45000.000000 |  
| 30 | 62500.000000 |  
| 50 | 60000.000000 |  
+-----+-----+  
4 rows in set (0.00 sec)
```

8. UPDATE Statement Modifies All Rows

```
mysql> UPDATE Employee  
-> SET Salary=70000;  
Query OK, 5 rows affected (0.01 sec)  
Rows matched: 6 Changed: 5 Warnings: 0  
  
mysql> UPDATE Employee  
-> SET Salary=70000  
-> WHERE EmpID=101;  
Query OK, 0 rows affected (0.00 sec)  
Rows matched: 1 Changed: 0 Warnings: 0
```

9. Optimize Multi-table Query

```
mysql> SELECT Name
      -> FROM Employee
      -> WHERE DeptID IN
      -> (
      -> SELECT DeptID
      -> FROM Department
      -> WHERE ManagerID IN
      -> (
      -> SELECT ManagerID
      -> FROM Manager
      -> )
      -> );
```

```
+-----+
| Name |
+-----+
| Raj  |
| Arun |
| Divya |
| Meena |
| Karan |
| John |
+-----+
```

6 rows in set (0.00 sec)

```
mysql> SELECT E.Name,
      -> D.DeptName,
      -> M.ManagerName
      -> FROM Employee E
      -> JOIN Department D
      -> ON E.DeptID=D.DeptID
      -> JOIN Manager M
      -> ON D.ManagerID=M.ManagerID;
```

```
+-----+-----+-----+
| Name | DeptName | ManagerName |
+-----+-----+-----+
| Raj  | IT       | Kumar       |
| Arun | IT       | Kumar       |
| Divya | HR       | Priya       |
| Meena | Finance  | Ravi        |
| Karan | Finance  | Ravi        |
| John | Marketing | Priya       |
+-----+-----+-----+
```

6 rows in set (0.00 sec)

Question 10: Relational Calculus Equivalent SQL

```
mysql> SELECT DISTINCT E.Name  
-> FROM Employee E  
-> WHERE EXISTS  
-> (  
-> SELECT *  
-> FROM Works W  
-> WHERE W.EmpID=E.EmpID  
-> );
```

```
+-----+  
| Name |  
+-----+  
| Raj  |  
| Arun |  
| Divya |  
| Meena |  
| Karan |  
+-----+
```

```
5 rows in set (0.02 sec)
```